# WEEK 4 REPORT

*Machine Learning Model Development and Evaluation*

## 1. Objective

The objective was to develop and evaluate a supervised machine-learning classifier using a publicly available dataset, select an appropriate algorithm, tune its hyperparameters with cross-validation, and interpret its predictive performance.

## 2. Dataset

The scikit-learn Wine Recognition dataset, associated with the UCI Machine Learning Repository, was used. It contains 178 wine observations, 13 numerical chemical predictors, and three target classes. The dataset is small and curated, making it suitable for a reproducible classification exercise.

## 3. Preprocessing and split

The target labels were mapped to class_0, class_1 and class_2. A stratified 75:25 train/test split with random_state=42 was used. Stratification preserves approximate class proportions. Random Forest does not require feature scaling, so raw numerical predictors were supplied to the estimator.

## 4. Algorithm selection

Random Forest was selected because it combines multiple decision trees, can capture nonlinear relationships and feature interactions, is relatively robust to feature scaling, and provides feature-importance estimates.

## 5. Hyperparameter tuning

GridSearchCV with 5-fold StratifiedKFold cross-validation optimized accuracy on the training set. The search considered n_estimators = 100 or 200; max_depth = None, 5 or 10; min_samples_split = 2 or 5; and min_samples_leaf = 1 or 2. Best parameters: {'model__max_depth': None, 'model__min_samples_leaf': 1, 'model__min_samples_split': 2, 'model__n_estimators': 200}. Best mean cross-validation accuracy: 0.9852.

## 6. Test evaluation

On the held-out test set, accuracy was 1.0000, macro precision was 1.0000, macro recall was 1.0000, and macro F1-score was 1.0000. Macro averaging gives each class equal weight.

## 7. Cross-validation

Five-fold stratified cross-validation of the selected estimator across the full dataset produced mean accuracy 0.9830 with standard deviation 0.0228. This is an additional stability estimate, not independent external validation.

## 8. Interpretation and limitations

The evaluation indicates how the tuned model performed on this particular stratified split. Random Forest can model nonlinear patterns, but it may be less interpretable than a simple linear model. The benchmark is small and curated; high scores should not be generalized automatically to real-world wine samples. Grid search only evaluates the parameter combinations specified.

## 9. Conclusion

A reproducible Random Forest classification workflow was implemented, including stratified splitting, cross-validated hyperparameter search, held-out metrics, and cross-validation. The notebook contains executable code to reproduce the analysis.

## 10. References

UCI Machine Learning Repository — Wine dataset (https://archive.ics.uci.edu/dataset/109/wine).

Scikit-learn documentation — load_wine, RandomForestClassifier, GridSearchCV, and model evaluation metrics.